

ARISTOTLE UNIVERSITY OF THESSALONIKI
FACULTY OF ENGINEERING
SCHOOL OF MECHANICAL ENGINEERING

Production Planning and Control

1st COURSE ASSIGNMENT

**OPTIMAL DESIGN OF A PRODUCT PRODUCTION AND
DISTRIBUTION NETWORK**

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Introduction

This assignment deals with the optimization of the distribution network of an industrial company with two production units in Greece (the Nea Santa and Komotini factories). The company wishes to create a distribution center in Germany to serve the agents and customers in that country. The company produces twelve (12) different household-appliance products, has selected eight (8) candidate locations for creating the distribution center (DC), and additionally the locations of the ten (10) Customer/Agent Zones (henceforth referred to as customers) are known. The following costs are given:

- the production cost of each product at each factory (€/unit),
- the installation cost at each candidate location (€),
- the handling cost at each candidate location (€/unit),
- the transport cost from the factories to the distribution centers (€/unit),
- the transport cost from the distribution centers to the customer zones (€/unit).

Additionally, the demand for each product from each customer zone (units) is known, as is the relative amount of storage capacity required to keep each product at the distribution centers (m^2/unit). The decision on which distribution center to select will be made using the criterion of the minimum total network cost. The tables are provided in the Appendix.

The problem will be solved with the help of the Python language using the GAMS library, gamspy, which specializes in optimization problems. The questions will be answered through the Mixed-Integer Linear Programming (MILP) method.

Question a

In the first question, it is required to determine the location of the distribution center that minimizes the cost and satisfies customer demand. To solve the problem, the mathematical model needs to be formulated. First the indices will be defined, then the parameters and variables, and finally the objective function of the problem.

Indices:

- i : Products,
- j : Factories,
- k : Candidate distribution centers,
- l : Customer zones.

Parameters:

- $c_{i,j}$: unit production cost of each product at each factory,
- b_k : installation cost at each candidate location,
- h_k : unit handling cost at each candidate location,
- $v_{i,k}$: storage capacity for one unit of product at the distribution center,

$CD_{j,k}$: unit transport cost of the products from the factory to the distribution center (cost to distribution center, CD),

$CC_{k,l}$: unit transport cost of the products from the distribution center to the customer/agent zone (cost to customer, CC),

$D_{i,l}$: demand for the products from the customer zone.

Variables:

$X_{ij,k}$: production quantity of a product at the factory that is transported to the distribution center (continuous, positive variable),

$W_{i,k,l}$: quantity of product exiting the distribution center for the customer zone (continuous, positive variable),

Y_k : decision variable (binary $\{0,1\}$); 1 if DC k opens, 0 otherwise,

Z : objective function.

Constraints:

Next, we define the constraints of the problem, which have not been mentioned so far. For the production capacity, no maximum or minimum production-volume constraint is given. The same holds for the capacity of the warehouses. The first constraint appears in the storage at the distribution centers. The maximum capacity of each center is 50,000 m². This is expressed mathematically by the following relation:

$$\sum(l) v_{i,k} \cdot W_{i,k,l} \leq 50000 \cdot y_k \quad \forall k$$

The variable W is used to denote the product that entered, was handled, and is ready to be exported. There would be no difference if variable X were used in its place, as it is an arbitrary choice. This holds because whatever quantity enters the distribution center (X_{ijk}), the same must exit, so that flow balance holds:

$$\sum(j) X_{ij,k} = \sum(l) W_{i,k,l} \quad \forall i,k$$

The total exiting product must equal the customer demand so that system optimization is achieved, therefore:

$$\sum(k) W_{i,k,l} = d_{i,l} \quad \forall i,l$$

The variable Y_k denotes whether the distribution center is selected or not. The constraint given by the assignment states that ultimately only one distribution center will be selected. This is translated mathematically as:

$$\sum(k) Y_k = 1$$

Finally, there is a constraint on the transport of products from the factories to the distribution centers:

$$\sum(i) X_{ij,k} \leq 90000 \cdot Y_k \quad \forall j,k$$

as well as from the distribution centers to the customer zones:

$$\sum(i) W_{i,k,l} \leq 90000 \cdot Y_k \quad \forall k,l$$

It is understood that the last two constraints will not be active, since they are “cut off” by the capacity constraint of the distribution centers. The big-M method appears to be used, so that a constraint exists without affecting the region of feasible solutions.

Objective Function:

To complete the mathematical definition of the problem, the quantity to be minimized must be defined so that the problem can be solved. The variable Z (the objective function) is the decision criterion for the location of the distribution center. With this variable, based on the total cost, we try to minimize the cost value so as to reach the optimal solution. Its mathematical definition is given below:

$$\min Z = \sum(i,j,k) c_{ij} \cdot x_{ij,k} + \sum(k) b_k \cdot y_k + \sum(i,k,l) h_k \cdot w_{i,k,l} + \sum(i,j,k) cd_{j,k} \cdot x_{ij,k} + \sum(i,k,l) CC_{k,l} \cdot w_{i,k,l}$$

Solution:

After a Python script was created, defining and appropriately formulating the indices, parameters, variables and objective function with the help of the gamspy library, we arrive at the optimal solution, i.e. the minimum value of the total cost. The total cost amounts to €4,412,384.92 and the city in which the distribution center will be created is “Erfurt”, as shown in Image 1. The results include both the quantity and the production factory of each product.

Image 1 also depicts four further columns: level, marginal, upper, lower and scale. The level shows the value of the variable Y_k , while the marginal expresses how much better the optimal solution is than the next best. That is, if the company could not select Erfurt, the next choice would have an increased cost of €96,605. The lower and upper denote the lowest and highest values that the variable Y_k could take. The scale is an internal solver variable and does not affect the results.

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===== Optimal Solution =====

Minimum Total Cost = 4412384.92 EUR

Selected Distribution Center:
      k level marginal lower upper scale
3 Erfurt   1.0  96605.0    0.0   1.0   1.0

Selected Distribution Center: Erfurt

```

Image 1

Question b

Next, besides the optimal location, the model also gives as output the production plan of the factories: in which factory each product will be produced and in what quantity. Since the quantities represent the total demand for the product, the most economical production unit is selected in combination with the most economical transport. Image 2 shows that ten of the twelve product codes will be produced at the Nea Santa factory, while only the last two codes will be produced at the Komotini factory, but in noticeably larger quantities.

In Image 2, column i shows the product, j the production factory, k the distribution center (which is the same in every case), and Level the production quantity in units.

It is obvious that codes M1 and JK1 should be produced at the Komotini factory, as their production costs are three and five times lower, respectively, than at Nea Santa. Meanwhile, the production of the remaining codes is more economical at the Nea Santa factory — by 30% up to 920% (!).

The last table the program produces is the table of demand for the products from each customer zone. This table, shown in Image 3, coincides with Table 4 given in the assignment. This is desirable and acceptable, because it indicates that demand is fully covered without any compromise, and thus an optimal solution is achieved rather than merely a feasible one.

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===== Factory -> Flows of the selected DC =====

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	i	j	k	level	marginal	lower	upper	scale
3	PLS 502	Nea Santa	Erfurt	3760.0	0.0	0.0	inf	1.0
19	PLS 502L	Nea Santa	Erfurt	4235.0	0.0	0.0	inf	1.0
35	PLS 602	Nea Santa	Erfurt	2709.0	0.0	0.0	inf	1.0
51	PLS 602S	Nea Santa	Erfurt	2082.0	0.0	0.0	inf	1.0
67	PLS 602G	Nea Santa	Erfurt	1475.0	0.0	0.0	inf	1.0
83	PLS 602SF	Nea Santa	Erfurt	2016.0	0.0	0.0	inf	1.0
99	PLS 702	Nea Santa	Erfurt	2341.0	0.0	0.0	inf	1.0
115	PLS 702S	Nea Santa	Erfurt	2078.0	0.0	0.0	inf	1.0
131	PLS 702SF	Nea Santa	Erfurt	1263.0	0.0	0.0	inf	1.0
147	PLS 103NF	Nea Santa	Erfurt	2379.0	0.0	0.0	inf	1.0
171	M1	Komotini	Erfurt	15718.0	0.0	0.0	inf	1.0
187	JK1	Komotini	Erfurt	38973.0	0.0	0.0	inf	1.0

Image 2

Image 3

Thus, the objective function is broken down as shown in Image 4. It is evident and logical that the largest cost is the production cost, followed by the transport cost from the factories to the distribution center, while the lowest cost is that of handling, which is approximately half of the installation cost.

```

===== Cost Breakdown =====

```

Production:	2,988,480.94 EUR
Transport Factory -> DC:	1,123,578.57 EUR
DC Handling:	48,523.81 EUR
Transport DC -> Customers:	155,196.60 EUR
DC Installation (fixed):	96,605.00 EUR

Total (manual check):	4,412,384.92 EUR
Total (solver):	4,412,384.92 EUR

Image 4

Question c

In this question, it is required to create four (4) additional scenarios — two of higher demand (high, higher) and two of lower demand (low, lower) than the expected one (basic), which is presented in Table 4 and was used in the previous part of the assignment. The objective is to find the distribution center that minimizes the objective function of the expected total cost, taking into account all possible scenarios of customer-zone demand.

Consequently, we have an expansion of the mathematical model that will be run to solve this problem, with the addition of the index

s : demand scenario,
of the parameters

ψ_s : probability of occurrence of the scenario,
 f_s : demand factor of the scenario.

Also, the index s (denoting the scenario) is added to the demand parameter, so it becomes $d_{i,l,s}$, while the variables X , W and Z now take the index s and become:

$X_{i,j,k,s}$

$W_{i,k,l,s}$

Z_s : objective function value for scenario s .

Alongside the indices, parameters and variables, the constraints of the problem also change, as they are affected by the demand in the scenarios. Accordingly, we have:

$$\begin{aligned}\sum(k) w_{s_{i,k,l,s}} &= d_{s_{i,l,s}} \quad \forall i,l,s \\ \sum(j) x_{s_{i,j,k,s}} &= \sum(l) w_{s_{i,k,l,s}} \quad \forall i,k,s \\ \sum(i) x_{s_{i,j,k,s}} &\leq 90000 \cdot y_k \quad \forall j,k,s \\ \sum(i) w_{s_{i,k,l,s}} &\leq 90000 \cdot y_k \quad \forall k,l,s \\ \sum(s) prob_s &= 1 \\ z_{expected} &= \sum(s) \psi_s \cdot z_s\end{aligned}$$

The remaining constraints need no modification. The last constraint is essentially the mathematical definition of the expected cost that is to be minimized. In other words, what needs to be minimized is not the cost in each individual case, but the sum of these costs multiplied by their probability of occurrence. This is done so that the company is covered in any case and is sure it made the most correct choice regardless of the outcome.

The scenarios are divided into optimistic (high, higher) and pessimistic (low, lower). Nevertheless, demand has been calculated with specialized tools (as stated in the assignment), which means that large deviations from the basic demand forecast are not expected. Consequently, the extreme cases (higher, lower) vary by 25% from the basic model (factors 1.25 and 0.75) with a 5% probability of occurring. At the same time, the more “conservative” cases (high, low) vary by 10% from the basic model (factors 1.1 and 0.9) with a 15% probability of occurring. In conclusion, the basic scenario has a 60% probability of occurring. All of this is made clearer in the image below.

```

# NEW PARAMETERS
prob = Parameter(m, name="prob", domain=[s])
prob.setRecords([
    ("lower", 0.05),
    ("low", 0.15),
    ("basic", 0.60),
    ("high", 0.15),
    ("higher", 0.05),
])

```

Image 5

```

scenario_factors = {
    "lower": 0.75,
    "low": 0.90,
    "basic": 1.00,
    "high": 1.10,
    "higher": 1.25,
}

```

Image 6

Consequently, to calculate the customer demand in each case, the basic demand is multiplied by the corresponding factor, and the demand for each case results as presented in the appendix.

Subsequently, the code must select the optimal solution and then continue solving the model with respect to the production of the products. For this reason, a loop is created that sets $Y_k=1$ for each $k = \{1, \dots, 8\}$, in order to find the minimum expected cost. This means the model computes z for each scenario s and for each candidate distribution center, in order to find the minimum expected cost and continue with it. It is thus shown that the optimal solution is again Erfurt, with an expected cost of €4,414,280.32.

The individual costs of the 5 scenarios are given below, as well as their “burden” (weighted contribution) on the final result.

```

=====
STOCHASTIC MODEL - Finding Optimal DC (k) across all scenarios
=====

Solving stochastic model for each DC candidate ...

DC Candidate      Expected Cost (EUR)
-----
Wurzburg          4,466,171.44
Hanover           4,545,183.64
Magdeburg         4,500,151.09
Erfurt            4,414,280.32
Ingolstadt        4,637,066.41
Bielefeld         4,482,299.40
Cologne           4,494,668.61
Munich            4,450,739.76

-----
* OPTIMAL DC -> Erfurt (4,414,280.32 EUR)

```

Image 7

From this point onward, the problem is solved with the Mixed-Integer Linear Programming model, given that the best choice is Erfurt. The model also produces the optimal production for each scenario of the optimal solution, so that the difference in the demand of each product is better understood.

COST PER SCENARIO	
Total Cost (EUR)	
Scenario	
lower	3,337,721.68
low	3,984,173.39
basic	4,412,384.92
high	4,850,251.05
higher	5,495,992.44

Image 8

WEIGHTED SCENARIO CONTRIBUTION			
Cost (EUR) Probability Weighted Cost (EUR)			
Scenario			
lower	3,337,721.68	0.05	166,886.08
low	3,984,173.39	0.15	597,626.01
basic	4,412,384.92	0.60	2,647,430.95
high	4,850,251.05	0.15	727,537.66
higher	5,495,992.44	0.05	274,799.62
Expected Total Cost = 4,414,280.32 EUR			

Image 9

PRODUCTION PER SCENARIO					
DC: Erfurt					
Scenario	lower	low	basic	high	higher
Product					
PLS 502	2823	3386	3760	4141	4703
PLS 502L	3179	3814	4235	4662	5298
PLS 602	2035	2441	2709	2985	3390
PLS 602S	1564	1875	2082	2295	2604
PLS 602G	1110	1331	1475	1629	1848
PLS 602SF	1516	1817	2016	2222	2524
PLS 702	1760	2109	2341	2582	2931
PLS 702S	1561	1873	2078	2291	2601
PLS 702SF	952	1141	1263	1393	1584
PLS 103NF	1788	2144	2379	2622	2977
M1	11791	14150	15718	17295	19652
JK1	29233	35078	38973	42878	48719

Image 10

Finally, the model produces the table showing the individual costs.

OUTPUT 5: INDIVIDUAL COST BREAKDOWN PER SCENARIO						
DC: Ερφούρτη						
Scenario	Production Cost (€)	Transport Factory→DC (€)	Handling at DC (€)	Transport DC→Customers (€)	Fixed Installation (€)	Total (€)
lower	2,245,010.96	843,223.68	36,417.57	116,464.47	96,605.00	3,337,721.68
low	2,692,480.22	1,011,663.96	43,691.63	139,732.58	96,605.00	3,984,173.39
basic	2,988,480.94	1,123,578.57	48,523.81	155,196.00	96,605.00	4,412,384.92
high	3,292,621.80	1,236,788.31	53,414.93	170,821.01	96,605.00	4,850,251.05
higher	3,739,557.40	1,405,074.57	60,682.23	194,073.24	96,605.00	5,495,992.44
Weighted Avg (Expected):						
	2,990,082.29	1,123,829.89	48,535.26	155,227.88	96,605.00	4,414,280.32

Image 11